COMPARATIVE ANALYSIS OF MUTUAL FUND SCHEMES

Portfolio Analysis Report - Individual Assignment

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1. Introduction

This report presents a comparative performance analysis of three open-ended equity Multi Cap mutual fund schemes over a common period of approximately five years: Nippon India Multi Cap Fund (the allotted scheme), and two peer schemes from the same category - ICICI Prudential Multicap Fund and Quant Active Fund. All three schemes invest across large-cap, mid-cap and small-cap stocks (minimum 25% each, per SEBI's Multi Cap category mandate) and share the same benchmark, allowing for a like-for-like comparison. Each scheme is first profiled and evaluated individually, followed by a comparative analysis, and an overall conclusion and recommendation.

2. Methodology & Assumptions

1. Period: September 2021 to August 2026 (~5 years). After aligning all four data series (3 schemes + benchmark) to their common trading dates, 1,204 daily return observations (~4.8 years) were used identically across all schemes and the benchmark.
2. Return frequency: Daily NAV-based returns for all three schemes and the benchmark, annualised using 252 trading days per year.
3. Benchmark: NIFTY500 Multicap 50:25:25 TRI (Total Return Index) - the common benchmark for all three schemes.
4. Risk-free rate: 6.95% - 10-Year India Government Security (G-Sec) yield.
5. Data source: NAV (all three schemes) and benchmark index data sourced from investing.com. Risk-free rate sourced from Trading Economics.

Formulas used

1. Annualised Return (CAGR) = (Latest Price / Oldest Price) ^ (Trading Days per Year / N) - 1
2. Annualised Risk = Daily Std. Dev. of Returns x SQRT (Trading Days per Year)
3. Beta = SLOPE (Fund daily returns, Benchmark daily returns)
4. Sharpe Ratio = (Annualised Return - Risk-Free Rate) / Annualised Risk
5. Treynor Ratio = (Annualised Return - Risk-Free Rate) / Beta
6. Jensen's Alpha = Annualised Return - [Risk-Free Rate + Beta x (Benchmark Return - Risk-Free Rate)]

3. Nippon India Multi Cap Fund (Allotted Scheme)

3.1 Scheme Profile

Fund House	Nippon India Mutual Fund
Category	Equity - Multi Cap Fund
Inception Date	28 March 2005
Fund Manager(s)	Sailesh Raj Bhan; Ashutosh Bhargava (Co-Fund Manager, since Sep 2021)
Benchmark	NIFTY500 Multicap 50:25:25 TRI
Approx. AUM (2026)	~Rs. 55,000 Cr
Investment Mandate	Minimum 25% each in large-cap, mid-cap and small-cap stocks

3.2 Performance Metrics

Metric	Nippon India Multi Cap Fund	NIFTY500 TRI (Benchmark)
Annualised Return	17.80%	11.47%
Annualised Risk (Std Dev)	14.74%	14.38%
Beta	0.97	1.00
Sharpe Ratio	0.74	0.31
Treynor Ratio	11.16%	N/A
Jensen's Alpha	6.46%	0.00%

3.3 Individual Interpretation

1. Nippon India Multi Cap Fund delivered an annualised return of 17.80% over the period analysed, with volatility of 14.74% - only marginally above the benchmark's 14.38%. Its beta of 0.97 shows the fund moves almost in line with the broader market (NIFTY500 TRI), i.e., it is neither materially more nor less volatile than the market as a whole.
2. The fund's Sharpe ratio of 0.74 means investors earned 0.74 units of excess return (over the risk-free rate) for every unit of total risk taken - comfortably ahead of the benchmark's Sharpe of 0.31, indicating strong risk-adjusted performance in absolute terms. The Treynor ratio of 11.16% (return earned per unit of systematic/market risk) further confirms this.
3. Most notably, the fund posted a Jensen's Alpha of +6.46%, meaning it generated 6.46 percentage points of return above what CAPM would predict given its beta and the benchmark's performance - strong evidence of genuine fund-manager skill (stock selection and/or market timing) rather than performance driven purely by higher risk-taking.

4. ICICI Prudential Multicap Fund (Peer 1)

4.1 Scheme Profile

Fund House	ICICI Prudential Mutual Fund
Category	Equity - Multi Cap Fund
Inception Date	1 October 1994
Fund Manager(s)	Sharmila D'Silva
Benchmark	NIFTY500 Multicap 50:25:25 TRI
Approx. AUM (2026)	~Rs. 15,000-18,000 Cr
Investment Mandate	Minimum 25% each in large-cap, mid-cap and small-cap stocks

4.2 Performance Metrics

Metric	ICICI Prudential Multicap Fund	NIFTY500 TRI (Benchmark)
Annualised Return	15.81%	11.47%
Annualised Risk (Std Dev)	13.74%	14.38%
Beta	0.91	1.00
Sharpe Ratio	0.64	0.31
Treynor Ratio	9.69%	N/A
Jensen's Alpha	4.73%	0.00%

4.3 Individual Interpretation

1. ICICI Prudential Multicap Fund delivered an annualised return of 15.81%, while carrying the lowest volatility among all three schemes analysed (13.74%) - even below the benchmark's 14.38%. Its beta of 0.91 indicates the fund is somewhat less sensitive to market movements than the broader index, consistent with a comparatively defensive positioning.
2. Its Sharpe ratio of 0.64 (against the benchmark's 0.31) shows the fund delivered more than double the risk-adjusted return of the market on a total-risk basis. The Treynor ratio of 9.69% confirms a strong reward per unit of systematic risk taken.
3. A positive Jensen's Alpha of 4.73% indicates the fund outperformed its CAPM-predicted return by 4.73 percentage points, reflecting genuine value added by the fund management team - achieved with the lowest risk profile of the three schemes analysed, making this fund particularly attractive for more risk-conscious investors.

5. Quant Active Fund (Peer 2)

5.1 Scheme Profile

Fund House	quant Mutual Fund
Category	Equity - Multi Cap Fund
Inception Date	February 2001
Fund Manager(s)	Sandeep Tandon, Ankit Pande and team (quant model-based approach)
Benchmark	NIFTY500 Multicap 50:25:25 TRI
Approx. AUM (2026)	~Rs. 7,700 Cr
Investment Mandate	Minimum 25% each in large-cap, mid-cap and small-cap stocks

5.2 Performance Metrics

Metric	Quant Active Fund	NIFTY500 TRI (Benchmark)
Annualised Return	11.81%	11.47%
Annualised Risk (Std Dev)	17.58%	14.38%
Beta	1.10	1.00
Sharpe Ratio	0.28	0.31
Treynor Ratio	4.43%	N/A
Jensen's Alpha	-0.10%	0.00%

5.3 Individual Interpretation

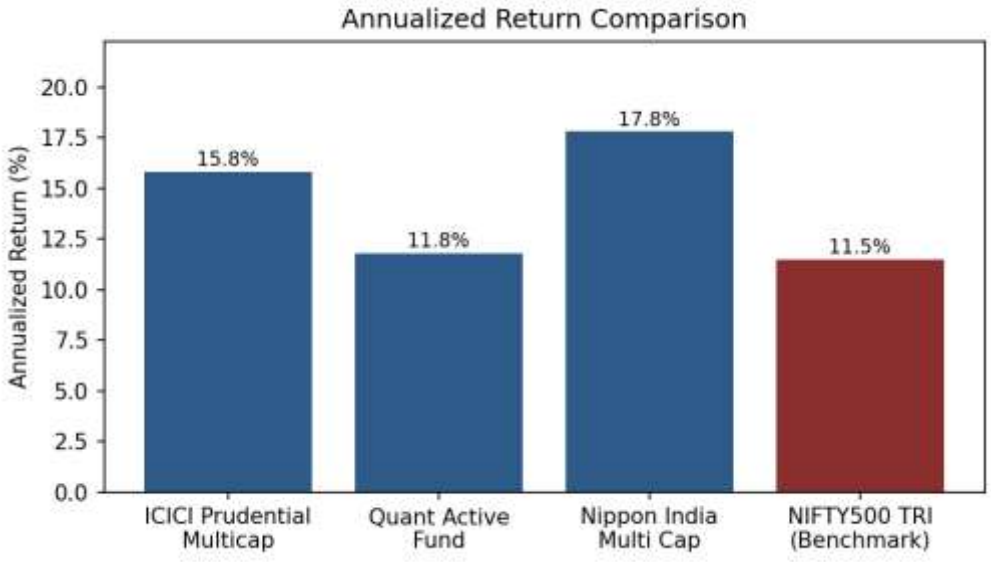
1. Quant Active Fund delivered an annualised return of 11.81%, only marginally ahead of the benchmark's 11.47% and the weakest of the three schemes analysed. This modest return came with the highest volatility of the group (17.58% std. dev.), well above both peer funds and the benchmark. Its beta of 1.10 confirms the fund is more aggressive than the market, amplifying market movements in both directions.
2. Because its return did not increase proportionately with its higher risk, the fund's Sharpe ratio of 0.28 actually falls below the benchmark's 0.31 - meaning that, on a total-risk-adjusted basis, an investor would have been better off simply holding the benchmark index. The Treynor ratio of 4.43% (versus 9.69%-11.16% for the peer funds) reinforces this finding.
3. Most tellingly, the fund's Jensen's Alpha is marginally negative (-0.10%), indicating its actual return was slightly below what CAPM would predict given its beta - i.e., essentially no risk-adjusted outperformance, and a small risk-adjusted shortfall, over the period analysed.

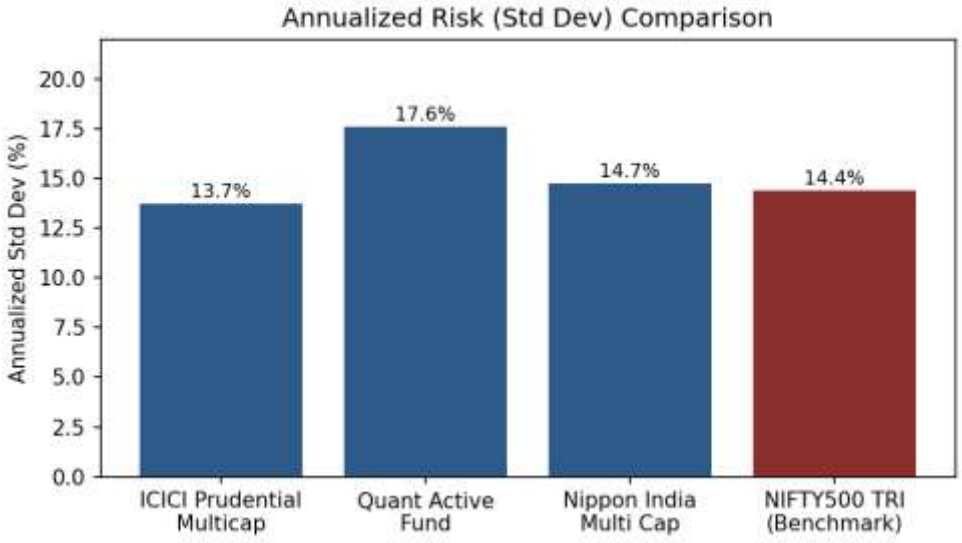
6. Comparative Analysis

6.1 Comparative Summary

Particulars	ICICI Prudential Multicap Fund	Quant Active Fund	Nippon India Multi Cap Fund	NIFTY500 TRI (Benchmark)
Annualised Return	15.81%	11.81%	17.80%	11.47%
Annualised Risk (Std Dev)	13.74%	17.58%	14.74%	14.38%
Beta	0.91	1.10	0.97	1.00
Sharpe Ratio	0.64	0.28	0.74	0.31
Treynor Ratio	9.69%	4.43%	11.16%	N/A
Jensen's Alpha	4.73%	-0.10%	6.46%	0.00%

6.2 Return and Risk Charts





6.3 Comparison Interpretation

On raw return, Nippon India Multi Cap Fund leads (17.80%), followed by ICICI Prudential Multicap Fund (15.81%); Quant Active Fund (11.81%) is barely ahead of the benchmark (11.47%). On risk, the ranking is reversed at the bottom: ICICI is the least volatile (13.74%), Nippon India is close to the benchmark (14.74% vs 14.38%), while Quant Active is distinctly the most volatile (17.58%).

Beta shows ICICI (0.91) and Nippon India (0.97) both positioned slightly more defensively than the market, while Quant Active (1.10) is more aggressive. This risk profile directly explains the ratio results: Nippon India posts the best Sharpe (0.74) and Treynor (11.16%) ratios, ICICI is a close second (Sharpe 0.64, Treynor 9.69%), and both comfortably beat the benchmark (Sharpe 0.31). Quant Active, despite similar or higher risk, lands below the benchmark on Sharpe (0.28) - its extra volatility was not rewarded with proportionately higher return.

Jensen's Alpha reinforces this same ordering: Nippon India (+6.46%) and ICICI (+4.73%) both show genuine, CAPM-adjusted outperformance, while Quant Active (-0.10%) shows essentially none. Across every measure examined, the ranking is consistent: Nippon India Multi Cap Fund > ICICI Prudential Multicap Fund > Quant Active Fund.

7. Overall Interpretation & Conclusion

Taken together, the individual scheme analysis and the comparative analysis point to the same conclusion. Nippon India Multi Cap Fund is the strongest performer of the three schemes over the ~5-year (aligned ~4.8-year) period studied - it delivered the highest absolute return while keeping risk close to the benchmark, and it is the only scheme to lead on every risk-adjusted measure examined (Sharpe ratio, Treynor ratio and Jensen's alpha). This combination indicates the outperformance is attributable to genuine fund-manager skill rather than simply taking on more risk.

ICICI Prudential Multicap Fund is a strong second choice: it could not quite match Nippon India's absolute return, but it achieved the lowest volatility of the three schemes while still comfortably beating the benchmark on every risk-adjusted measure. It is well suited to investors who prioritise capital stability alongside genuine outperformance.

Quant Active Fund is the weakest of the three on a risk-adjusted basis. Its higher beta and volatility were not compensated by a proportionately higher return, resulting in the lowest Sharpe and Treynor ratios of the group and a marginally negative Jensen's Alpha - indicating its performance was essentially indistinguishable from (or fractionally below) what CAPM would predict given its risk level.

Final Ranking

1. 1st - Nippon India Multi Cap Fund: Highest return (17.80%) and best risk-adjusted performance on every measure (Sharpe 0.74, Treynor 11.16%, Jensen's Alpha 6.46%).
2. 2nd - ICICI Prudential Multicap Fund: Lowest risk (13.74%) with strong, consistent risk-adjusted outperformance (Sharpe 0.64, Treynor 9.69%, Jensen's Alpha 4.73%).
3. 3rd - Quant Active Fund: Highest risk (17.58%) without commensurate return; weakest Sharpe (0.28), Treynor (4.43%) and a marginally negative Jensen's Alpha (-0.10%).

Recommendation

Nippon India Multi Cap Fund is recommended as the best-performing scheme among the three over the analysis period, on both an absolute and a risk-adjusted basis. ICICI Prudential Multicap Fund is a credible, lower-risk alternative for more conservative investors. Quant Active Fund, despite sharing the same category mandate, did not adequately compensate investors for the additional volatility taken on during this period and would require a longer track record or improved risk management to be preferred over its peers.